

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459598.9815 +/- 0.0055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.053
Transit depth (Rp/Rs)^2	3.29 +/- 1.94 [%]
Orbital Inclination (inc)	86.47 +/- 1.79
Airmass coefficient 1 (a1)	0.153 +/- 0.095
Airmass coefficient 2 (a2)	0.72 +/- 0.17
Scatter in the residuals of the lightcurve fit is	55.79 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.0
Transit Duration (day)	0.059 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.181 ± 0.053 vs 0.1326 (deviation 0.48σ)
Duration fit vs published	0.059 d vs 0.1238 d (deviation 1.61σ)
Mid-time O–C	-9.9 min
Depth SNR	1.8
Residual scatter	55.79 %

Light curve

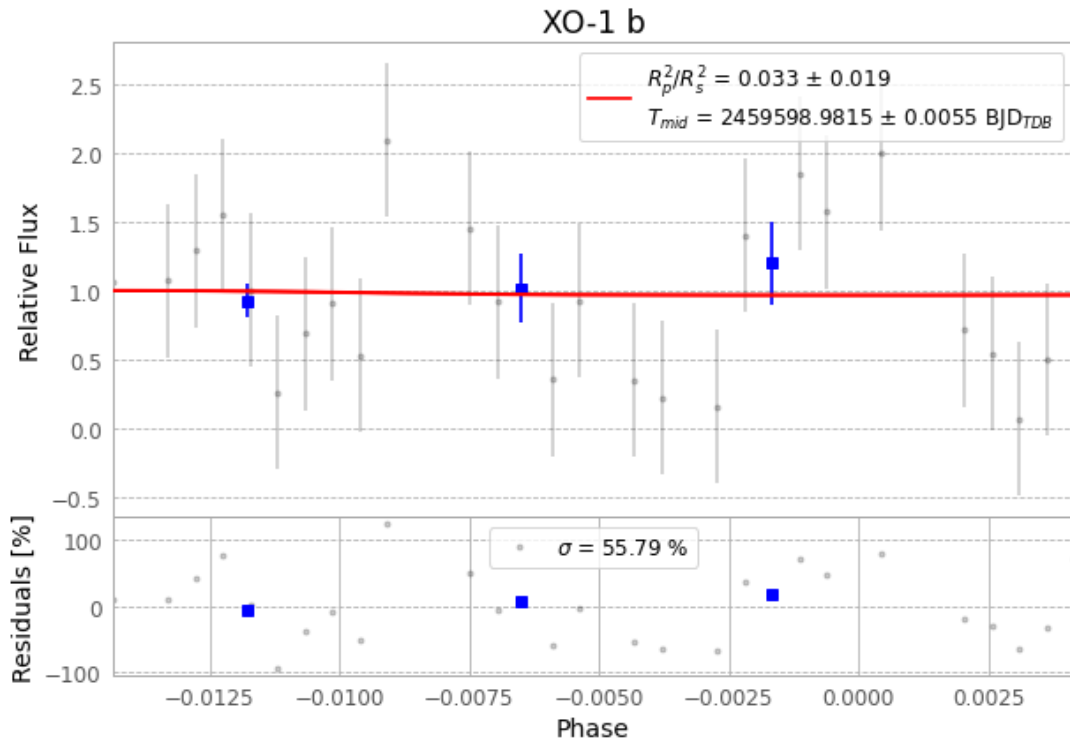


Figure 1 — FinalLightCurve_XO-1 b_2022-01-19.png (SHA-256 c02d59e1ee74b45b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [604, 172], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 825e838b03467f86...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

37 FITS frames (24 MB), XO-1220119101212.FITS → XO-1220119120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-220119.log (b67865d78cda...), FinalLightCurve_XO-1 b_2022-01-19.png (c02d59e1ee74...), FinalLightCurve_XO-1 b_2022-01-19.csv (991865980a3b...)

Compute resources

Wall time 160.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:11Z.